Climate Stress and Ecosystem Monitoring

GS III

- Climate events such as **El Niño** can influence rainfall, temperature and water availability across different regions. Their effects extend beyond weather conditions because changes in moisture and heat can affect vegetation, agriculture, forests and the survival of different species.

- Ecosystems respond differently to prolonged dry conditions. Forests, grasslands and wetlands have different water requirements and recovery capacities. Even areas receiving similar rainfall may experience different levels of **ecological stress** because soil moisture, vegetation and groundwater conditions vary.

- **Satellite observations** can track changes in vegetation, water bodies, snow cover and other environmental conditions across large areas. However, ground-level measurements remain necessary to interpret these changes accurately. Combining both methods helps identify ecological damage that ordinary weather forecasts may overlook.

- Continuous ecosystem monitoring can provide **early warnings** of environmental deterioration. Detecting declining vegetation health, shrinking wetlands or changing snow conditions allows authorities to plan conservation and adaptation measures. Effective monitoring requires cooperation among scientific institutions, government departments and local observers.

Climate preparedness requires monitoring how ecosystems respond to changing conditions, combining satellite information, field observations and coordinated action to identify emerging environmental risks.